# A2: Develop AI Powered Solutions using N-Aible AI business simulator.

Start Assignment

- Due Mar 24 by 11:59pm
- Points 100
- Submitting a file upload
- Available Mar 10 at 12am - Mar 26 at 11:59pm

**You will be using n-gage by n-aible to play this business case simulation.**

**Click on the below link to go to the business case simulator:**

https://www.stag.n-aible.com/invite/A2eoDlwa0BFt5bgN-LUJ_gsns9AtlpID9K9VmigNfuU (https://www.stag.n-aible.com/invite/A2eoDlwa0BFt5bgN-LUJ_gsns9AtlpID9K9VmigNfuU)

**Team Simulation:** Text Analytics for Strategic Brand Intelligence :: NIKE

**Executive Summary:**

In today's digital economy, organizations generate and are discussed through vast amounts of unstructured text data — customer reviews, social media posts, call center transcripts, product feedback, and competitor commentary. Extracting strategic insight from this data is now a core business competency.

This team-based simulation places students in the role of a Data & AI Consulting Team engaged by an external client. Students will:

1. Collect real-world unstructured text data

2. Process and analyze the data using Natural Language Processing (NLP) in R

3. Generate executive-level business insights

4. Deliver findings in a client-facing presentation

The simulation mirrors real consulting engagements, requiring technical rigor, business framing, and communication clarity.

**Real-World Constraints**

Students must account for:

• Sampling bias

• Fake review detection limitations

• Lexicon bias in sentiment analysis

• Ethical data usage

• Overfitting in classification models

Learning Objectives

By the end of this simulation, students will:

• Design an end-to-end NLP pipeline in R

• Translate unstructured data into strategic business decisions

• Bridge technical and executive communication

• Defend analytics under executive questioning

• Work as a client-facing analytics consulting team

Closing Statement to Students:

This simulation is not about producing elegant code.

It is about answering this question:

“If this were your consulting engagement, would the client pay for this analysis?”

Your success will be measured not by model complexity —

but by clarity, defensibility, and business impact.

## *External Client:: Nike*

## *Executive Context:*

Nike’s leadership has observed:

• Increasing online conversation around sustainability

• Mixed sentiment about pricing

• Strong engagement with new product drops

• Growing competition from challenger brands

The Chief Marketing Officer and VP of Consumer Insights have commissioned your team to

answer:

“What are customers really saying about Nike — and how should we respond strategically?”

Your consulting team must analyze large-scale unstructured text data to uncover brand perception drivers, product risks, and growth opportunities.

## 🎯 Strategic Business Questions

Your analysis must address at least four of the following:

Brand Perception

• How is Nike emotionally positioned in online conversations?

• What themes are most associated with brand loyalty?

Product Strategy

• What complaints drive negative reviews of specific product lines?

• Are there recurring quality or sizing issues?

Sustainability Narrative

• Is Nike's sustainability messaging resonating?

• Is "greenwashing" language appearing?

Pricing Sensitivity

• Are customers increasingly describing Nike as "overpriced"?

• Does negative sentiment cluster around premium lines?

Competitive Positioning

Benchmark against:

• Adidas

• Under Armour

What differentiates brand language?

## Required Data Sources (Student-Collected)::

Teams must collect text from at least two of the following:

• Nike product reviews (Nike.com or retail sites)

• X (Twitter) mentions

• Reddit sneaker communities

• YouTube comments on sneaker reviews

• App store reviews for Nike Training Club

• News headlines discussing Nike

**Required NLP Techniques in R. Students must implement:**

1.Text Preprocessing

• Tokenization

• Stopword removal

• Lemmatization or stemming

• N-grams

• Document-Term Matrix

2. Sentiment Analysis

• NRC or Bing lexicon

• Emotion breakdown (joy, anger, trust, anticipation)

• Sentiment by product line or competitor

3.Topic Modeling (LDA)

Identify:

• Core complaint themes

• Product differentiation themes

• Emerging brand risks

4.Business-Driven Text Analytics (Choose 2+)

• TF-IDF brand differentiation

• Aspect-based sentiment

• Text clustering

• Keyword co-occurrence network

• Classification model predicting low-rating reviews

***Expected Business Insight Types::***

Students must translate findings into strategic recommendations such as:

1. Brand Strategy

• Emotional drivers of loyalty

• Trust vs. price tension

• Athlete endorsements impact

2. Product Operations

• Material quality issues

• Fit/sizing inconsistencies

• Supply chain delays

3. Market Risk

• Backlash patterns

• Cultural controversy signals

• Sustainability skepticism

4. Growth Opportunities

• High-engagement product lines

• Trending design attributes

• Influencer language patterns

## Final Deliverables (What you must submit)::

1. Technical Appendix (R Markdown - including R code and R outputs in the Markdown file)

• Reproducible code

• Modeling details

• Limitations

2. Executive Presentation (10-12 Slides)

Audience:

• CMO

• VP Product

• Director of Brand Strategy

Must include:

• 5–8 strategic insights

• Clear executive recommendations

• Business impact framing

• Visual storytelling

3. 10-Minute Client VIDEO Presentation - Upload to YouTube.com as Unlisted.

Students must defend:

• Modeling choices

• Sampling bias

• Lexicon limitations

• Strategic implications

4. Screenshot of your n-gage by n-aible simulation final screen - with the automatic AI grading. We want to verify that you've used the Client's platform to play this simulation.

IMPORTANT: You must upload all 4 items above to receive a passing grade. Failure to upload any of the above will result in an F grade.

🔥 ***Optional Advanced Challenge:***

Build a classification model to predict whether a product review will be:

• 1–2 stars

• 3 stars

• 4–5 stars

Then identify language signals most predictive of churn risk.

| Nike Business Case simulation |
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| Criteria | Ratings | | | | | Pts |
|---|---|---|---|---|---|---|
| Business Framing & Problem Definition<br>What Professors Look For<br>• Did the team define a decision the client must make?<br>• Did they frame hypotheses?<br>• Is there financial or operational impact discussed? | 20 pts<br>Exceeds expectations<br>Clear articulation of client problem, decision context, and strategic stakes. Defines measurable business questions. Aligns analytics with executive priorities. | 17 pts<br>Above average<br>Business problem stated clearly but partially operationalized. Some alignment to strategy. | 14 pts<br>Average<br>Problem described at high level; weak linkage to decision-making. | 12 pts<br>Below average<br>Academic framing only; no clear business application. | 0 pts<br>No Marks<br>Missing | 20 pts |
| Text Preprocessing & Feature Engineering<br>Technical Expectations<br>• Proper token filtering<br>• Handling negation (e.g., “not clean”)<br>• Meaningful n-gram usage<br>• Clear reproducible R workflow | 20 pts<br>Exceeds expectations<br>Robust preprocessing pipeline (tokenization, stopwords, lemmatization/stemming, n-grams). Clean DTM. Justified choices. | 17 pts<br>Above average<br>Standard cleaning performed but limited sophistication. | 14 pts<br>Average<br>Basic tokenization only; minimal cleaning rigor. | 12 pts<br>Below average<br>Poor preprocessing; unreliable outputs. | 0 pts<br>No Marks<br>The code is missing, or the code producess errors that cannot be easily identified and remediated. | 20 pts |
| NLP Methodology & Modeling Depth<br>Required Techniques<br>• Sentiment | 20 pts<br>Exceeds expectations<br>Multiple advanced NLP techniques (sentiment + LDA + TF-IDF + classification/clustering). Methods appropriate | 17 pts<br>Above average<br>Required techniques implemented | 14 pts<br>Average<br>Methods applied mechanically | 12 pts<br>Below average<br>Incorrect or inappropriate modeling. | 0 pts<br>No Marks<br>Missing and/or errors. | 20 pts |

| Criteria | Ratings | | | | | Pts |
|---|---|---|---|---|---|---|
| analysis<br>• Topic modeling (LDA)<br>• At least two additional analytical methods | and justified. Model tuning discussed. | correctly but limited depth. | with limited interpretation. | | | |
| Insight Generation & Business Translation | **20 pts**<br>**Exceeds expectations**<br>Clear, actionable, strategic insights. Quantified implications. Direct linkage to client decisions. Goes beyond “what” to “so what.” | **17 pts**<br>**Above average**<br>Solid findings but limited executive framing. | **14 pts**<br>**Average**<br>Descriptive reporting without strategic recommendations. | **12 pts**<br>**Below average**<br>Technical outputs presented without interpretation. | **0 pts**<br>**No Marks**<br>Missing or very limited | 20 pts |
| Visualization & Communication Quality<br>Expectations<br>• No raw R console output in slides<br>• Clear headlines on each slide<br>• Visuals answer business questions | **20 pts**<br>**Exceeds expectations**<br>Executive-ready visuals. Clean, labeled, decision-focused. Story flows logically. | **17 pts**<br>**Above average**<br>Clear visuals but minor clutter or technical overload. | **14 pts**<br>**Average**<br>Charts difficult to interpret; academic formatting. | **12 pts**<br>**Below average**<br>Poor visualization or unreadable outputs. | **0 pts**<br>**No Marks**<br>Missing / poor quality | 20 pts |
| | | | | | | Total Points: 100 |